Logistics Operations Performance Analysis & ML-Based Predictions

End-to-End Analytics using SQL, Python & Power BI

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Business Problem

Logistics operations generate large volumes of shipment data across routes, carriers, customers, and suppliers. However, decision-makers often lack consolidated visibility into:

- Shipment flow and delays
- Route and carrier performance
- Cost behavior and risk exposure
- This limits proactive decision-making and operational optimization.

Objective

- Analyze shipment and transportation data
- Monitor logistics KPIs
- Identify high-risk routes and shipments
- Enable data-driven operational decisions

Data Source & Scope

Dataset: Global Ocean Freight Shipments Dataset

Source: Experience-based synthetic data modeled on industry practices and public domain knowledge.

Volume: 60,000 ocean freight shipments

Time Period: 2024–2025

Carriers: MSC, COSCO, OOCL, ONE, Evergreen

Trade Lanes: Asia–Europe, US–Asia, ME routes

Scope Definition

- Historical shipment-level analysis
- Descriptive & diagnostic analytics
- No real-time or predictive planning systems

Methodology

Data Ingestion

Raw CSV data loaded into SQL database using Python Engine

1

Data Processing & ML (Python)

Data cleaning and validation
Feature engineering:

- Delay days & delay flag
- Shipment month, season
- Trade lanes

Modeling & Visualization (Power Bi)

- DAX
- New Measure & New Column

3

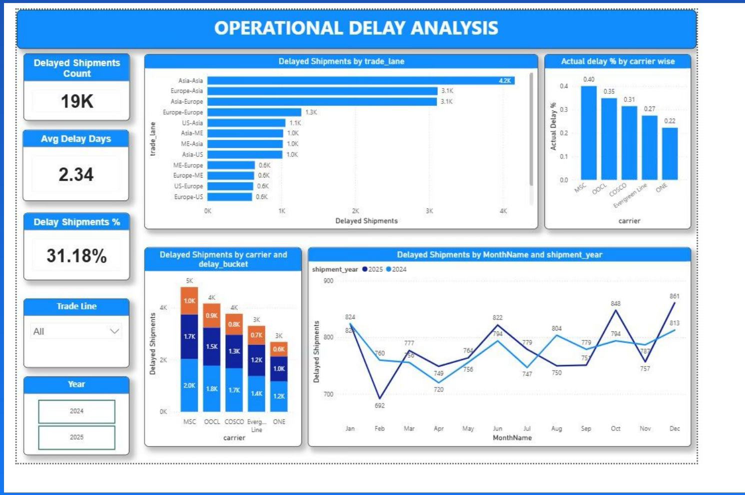
Operational Performance Analysis

The descriptive dashboards provide:

- Shipment volume trends by month
- Delay percentage across carriers
- High-traffic and high-delay trade lanes
- Customer and supplier contribution

Key Insight

Operational delays are lane- and carrier-dependent, with clear concentration in specific routes and seasons.



Shipment Delay Prediction

Problem:

Predict whether a shipment will be delayed

Models Used:

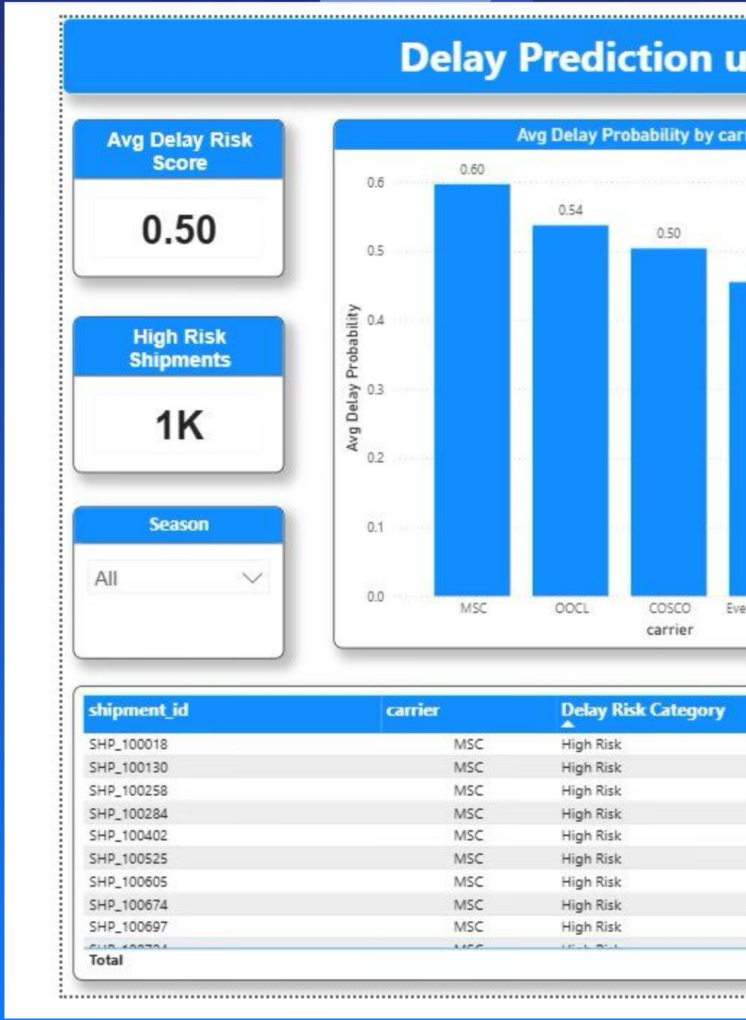
- Logistic Regression (baseline)
- Random Forest (final model)

Output:

- Delay probability score (0–1)
- **Risk categorization:**
  - High Risk
  - Medium Risk
  - Low Risk

Business Value

Enables early identification of high-risk shipments and proactive intervention.



Freight Cost Prediction

Objective:

Estimate base freight cost for planning and budgeting

Models Used:

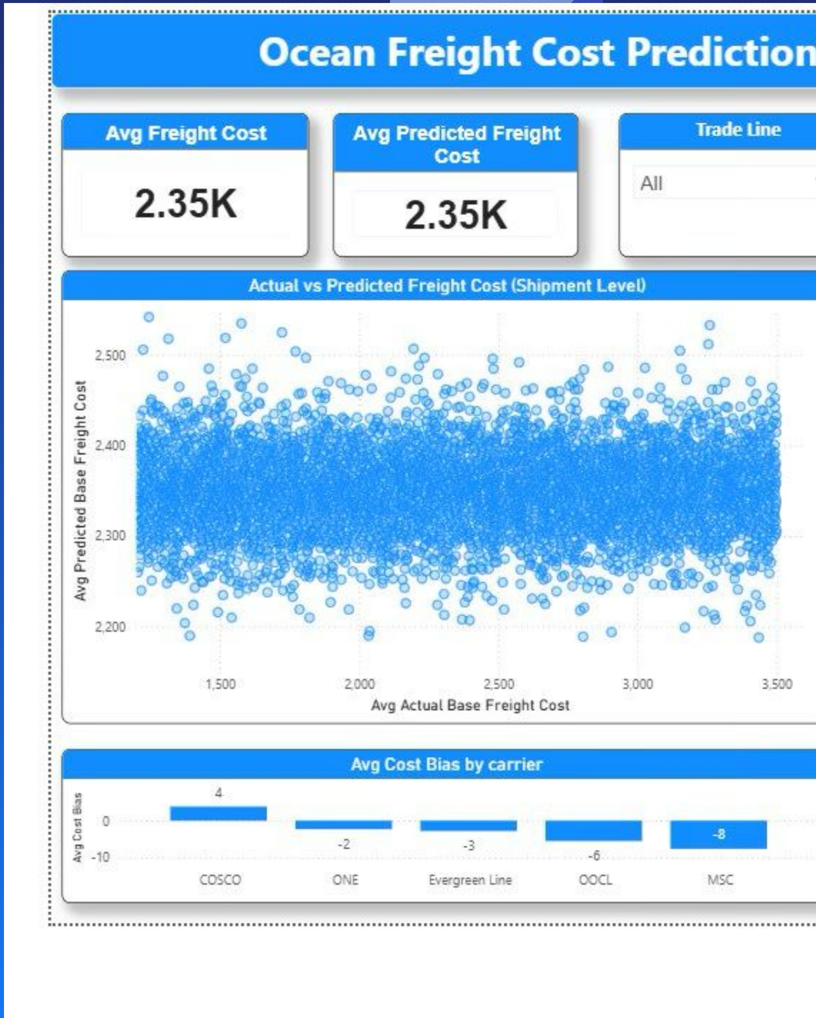
- Linear Regression (baseline)
- Random Forest Regressor (final)

Evaluation Outcome:

- Stable predictions
- Low bias
- Suitable for cost planning (not pricing automation)

Key Insight

Cost behavior varies more by route and carrier than by season alone.



Conclusion

This project delivers an end-to-end logistics analytics solution using:

- SQL for database storing datas.
- Python for data processing and feature creation
- Machine learning for delay and cost prediction
- Power BI for business-ready dashboards

Business Value:

- Improved operational visibility
- SLA negotiations
- Risk-based shipment prioritization
- Which places we should use buffer timing
- Detention and demurrage cost planning
- Better freight cost planning
- Clear separation of descriptive and predictive analytics

Thank you